LearnFlix

Project Proposal

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LearnFlix

Comprehensive Learning Management System

Abstract:

LearnFlix is a self-hosted learning management system designed for institutional digital education. The platform features three role-based dashboards: Student (course enrollment, progress tracking, AI assistant, notes, discussions, certificates), Teacher (content management, assignment/quiz creation, student reviews), and Admin (user management, permissions). Access is restricted to enrolled students only. The system integrates essential educational tools to streamline content delivery and learning experience management.

1. Introduction:

Students now rely more on digital sources of learning than traditional sources, and with ever-changing times, the need for flexible schedules, remote learning, and better online learning collaboration with instructors is emerging. Also, the need for access to new topics and new research in interactive video forms is emerging. All this is covered by Learning Management Systems, which are currently mostly proprietary solutions. We aim to create something for Pakistan, where the goal is to make this software system open-source and make access to course creation and hosting easy.

What sets Learnflix apart is its integration of "Smart Help." Recognizing that students often struggle in isolation, the platform embeds an AI Teaching Assistant powered by Large Language Models (LLMs) to provide instant glossary definitions, summary notes, and concept clarification without the student ever leaving the learning environment.

The frontend of the application will be made in react, the backend will be in django. Our project combines Django's robust, secure backend capabilities with React's dynamic, interactive frontend library.

Problem Statement:

Educational institutions lack an accessible, open-source learning management system that unifies content delivery, automated assessment, AI-powered student support, and analytics in a single self-hosted platform.

2. Functionalities:

Our project's objectives include:

- Developing a comprehensive Learning Management System for the web that combines content delivery, assessment, and interaction.
- Providing an intuitive course interface with support for modular content organization that adapts to different subject structures. Also, the Student Side is interactive with reviews and progress tracking with smart AI help and smart course recommendation.
- Enabling real-time student-instructor interaction through discussion forums and announcements to improve engagement.
- Integrating video hosting using YouTube API and PDF resource embedding to streamline access to learning materials.
- Implementing automated grading for quizzes and assignments to provide feedback to students.
- Offering personalized dashboards for both students and instructors to track grades, progress, and course completion.
- Providing analytics and visualizations to help students know where they stand and to assist teachers to know about content engagement.

3. Tech Stack:

Our project mainly focuses on developing a comprehensive Learning Management System for the web that integrates content delivery, assessment, and interaction into a single platform. For the frontend, we are using **React.js**, and for the backend, we are using the Django framework. Artificial Intelligence and automation will form a key part of the assessment and learning module. The system will support automated grading for multiple-choice quizzes and standardized questions. For analytics, libraries such as **Chart.js** will be adopted to display enrollment statistics, grade distributions, and completion rates in a clear and interactive manner for instructors. We would use 3rd party AI chat providers like **OpenAI**, and if we get time, we may try to use some model locally, via **LMStudio** to give better service to students. There is no offline synchronization for students (students right now cannot download resources like video lectures and other things). For deployment, we would use Docker to create a **Docker** image for easy hosting by people.

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4. Initial Study and Work Done so Far:

As far as our research is concerned, many platforms already exist to facilitate online learning. Platforms like edX, Coursera, and Canvas focus on high-scale university partnerships, while tools like Google Classroom focus on basic file management. However, many existing solutions are either too complex for a single class setup or too simple to offer robust analytics and structured course progression

Studies show that a unified interface significantly reduces the cognitive load on students. A recent study published in MDPI highlights that adaptive, centralized learning environments help manage "intrinsic" and "extraneous" cognitive load. This allows the students to focus better on core learning tasks rather than navigating complex systems (Alqurashi et al., 2025). Other research confirms that immediate feedback is critical for student retention. According to a 2025 study in Assessment & Evaluation in Higher Education, timely, personalized feedback significantly boosts student motivation and engagement, preventing them from feeling "lost" in the coursework (Hill & West, 2025). Furthermore, the integration of social features like discussion forums is vital for reducing isolation

Our aim is to integrate these fragmented elements: video lectures, resources, quizzes, grading, and analytics into one cohesive web application. So far, we have also reviewed existing LMS architectures..

References:

1. **Immediate Feedback and Student Motivation**

   Reference: Hill, J., & West, H. (2025). The impact of timely formative feedback on university student motivation. *Assessment & Evaluation in Higher Education*.

   Link: https://www.tandfonline.com/doi/full/10.1080/02602938.2025.2449891

2. **Social Interaction and Reducing Isolation**

   Reference: Müller, C., & Baumeister, V. (2025). Learning during COVID-19: (Too) isolated yet successful? *Frontiers in Education, 10*.

   Link: https://www.frontiersin.org/journals/education/articles/10.3389/feduc.2025.1549202/full

3. **Cognitive Load and Adaptive Interfaces**

   Reference: Alqurashi, E., & Alshehri, A. (2025). Challenging cognitive load theory: The role of educational neuroscience and artificial intelligence in redefining learning efficacy. *Brain Sciences*.

   Link: https://www.mdpi.com/2076-3425/15/2/203